

# DCEG Statistical Genetics Workshop

## Basic GWAS Analysis

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*September 16<sup>th</sup>, 2026*

# Outline

- Genetic epidemiology Intro
- Motivation for genetic studies
- Quality control/assessment

Session #1

A horizontal arrow points from the Session #1 box to the first list of topics.

- Statistical analysis
- Visualization and quality assessment
- Meta-analysis and imputation
- Annotation & reporting

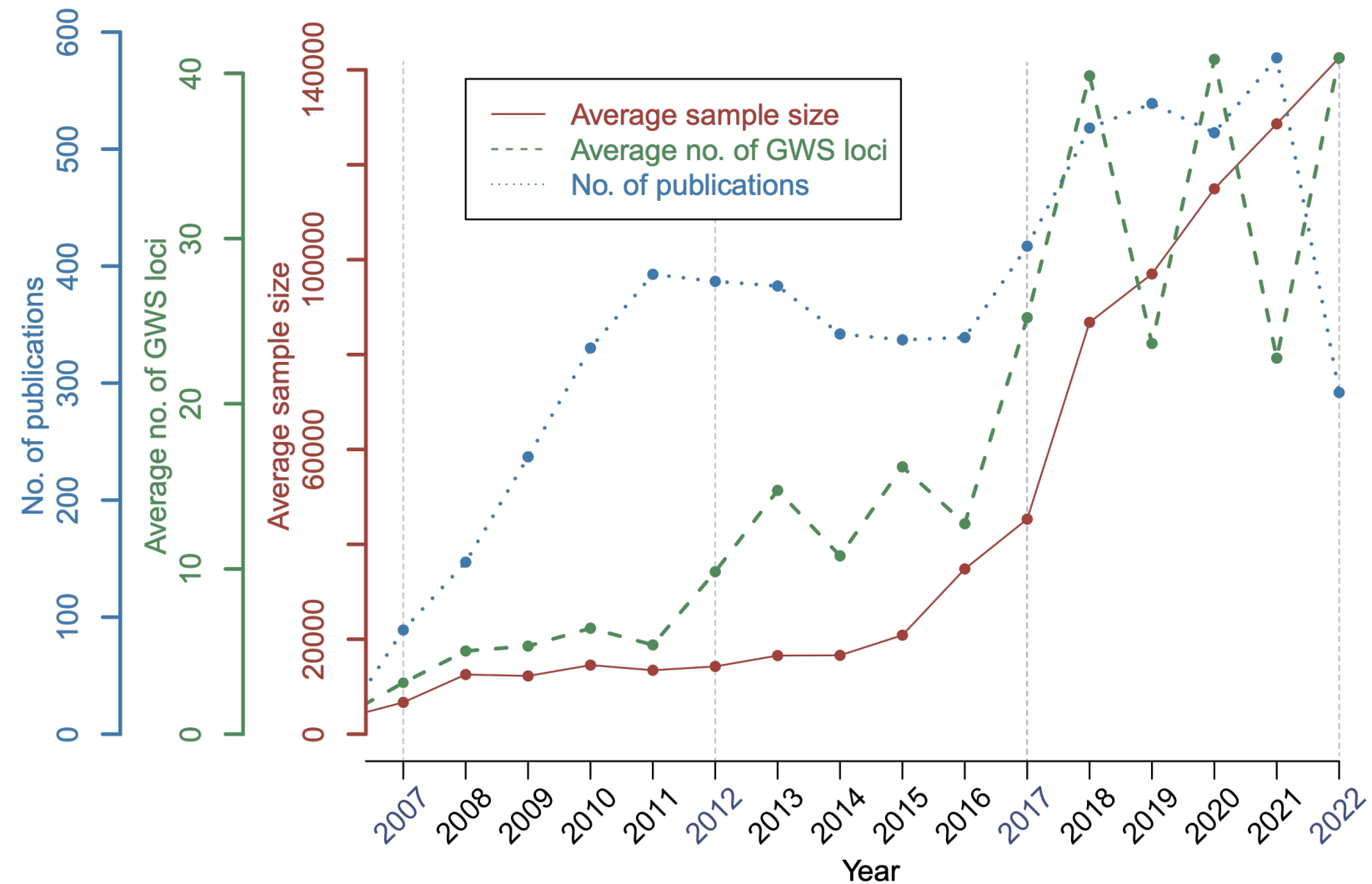
Session #2

A horizontal arrow points from the Session #2 box to the second list of topics.

# Quick GWAS Reminders

Sample  
Size:  
Bigger is  
Better

But?



**Figure 1. Average sample size and average number of genome-wide significant (GWS) loci per publication for each year during the 15 years history of GWAS discoveries**

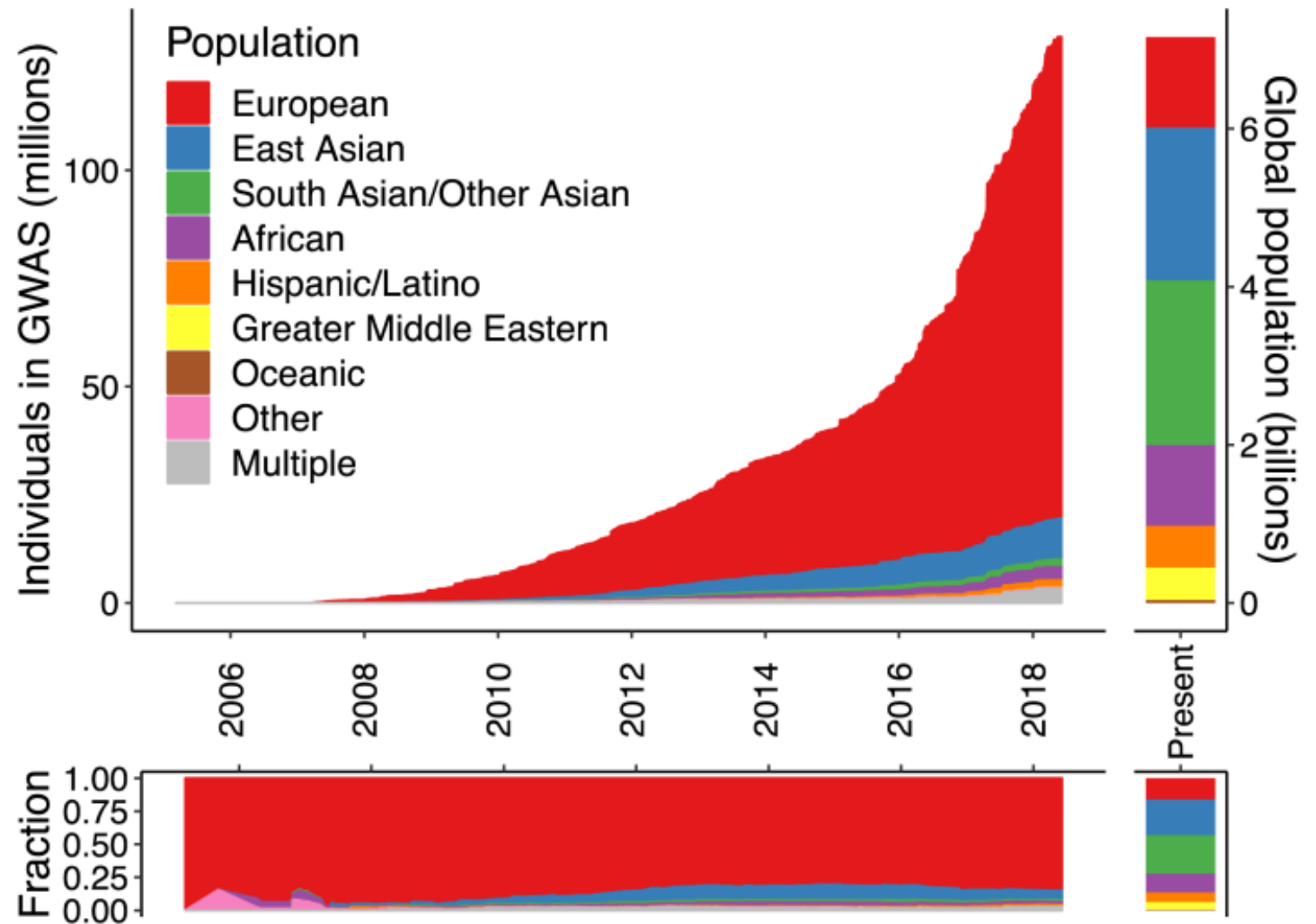
The data were extracted from 5,771 GWAS publications that used a genome-wide genotyping array and shared their summary statistics on GWAS Catalog before November 8, 2022.

# GWAS results are not always transferable across populations

- Genetic differences affect the distribution of allelic variants between groups
- Environmental differences can alter their effects within groups on traits and diseases
- The transferability of GWAS findings, including genetic associations and GWAS-derived polygenic predictors, depends on many factors such as:
  - Allele frequency differences
  - Linkage disequilibrium differences
  - Gene-by-gene interaction
  - Gene-by-environment interaction

There is a lack  
of diversity in  
GWAS

a.k.a. Euro-centric

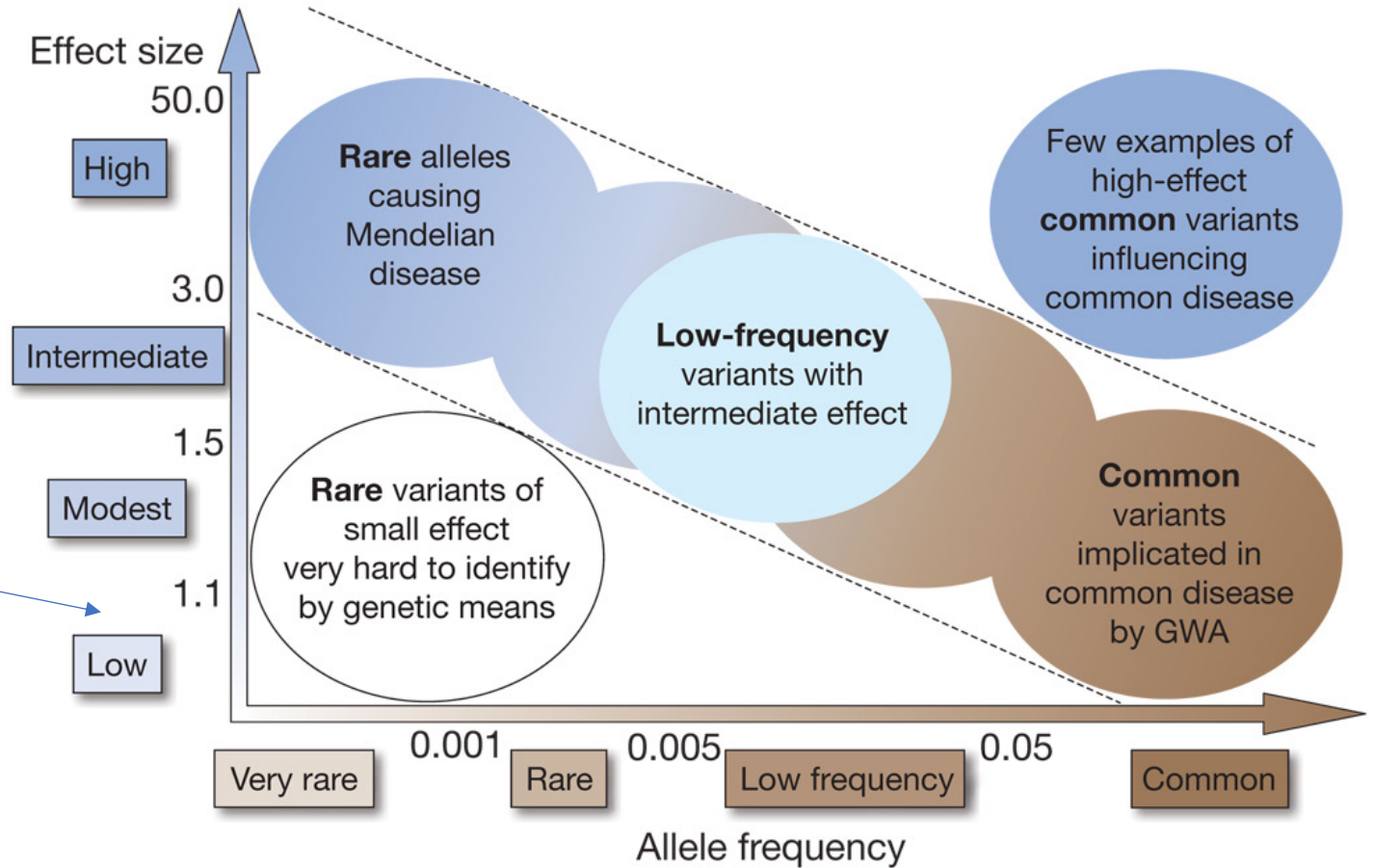


**Figure 1. Ancestry of GWAS participants over time compared to the global population.** Cumulative data as reported by the GWAS catalog<sup>76</sup>. Individuals whose ancestry is “not reported” are not shown.

# GWAS are mostly used for common variants

GWAS power calculations are a function of effect size, allele frequency, and sample size

Common variants have smaller effect sizes, which means large sample sizes are necessary to detect them!



Stay tuned for session #7 Rare Variant Analysis

# How to Conduct a GWAS?

# General GWAS Steps

- **Data management, quality control, quality assessment**
- **Association test**
  - Test each genetic variant for statistical association with the phenotype of interest
  - Adjust for covariates that could confound the association
  - Control the Type I error rate across the huge number of tests
- **Results visualization**
  - Q-Q plots, Manhattan plots, regional plots
- **Results annotation**
  - Variants to genes, regulatory elements, and biological function
- **Post-GWAS follow-up** → More to come in future sessions!

# Linear Regression

$$Y_i = X_i\alpha + G_i\beta + \epsilon_i$$

Assume  $Y_i$ s are independent given  $X_i, G_i$ .

$Y_i$ : phenotype vector for the  $i$ th individual

$X_i$ : covariates matrix for the  $i$ th individual

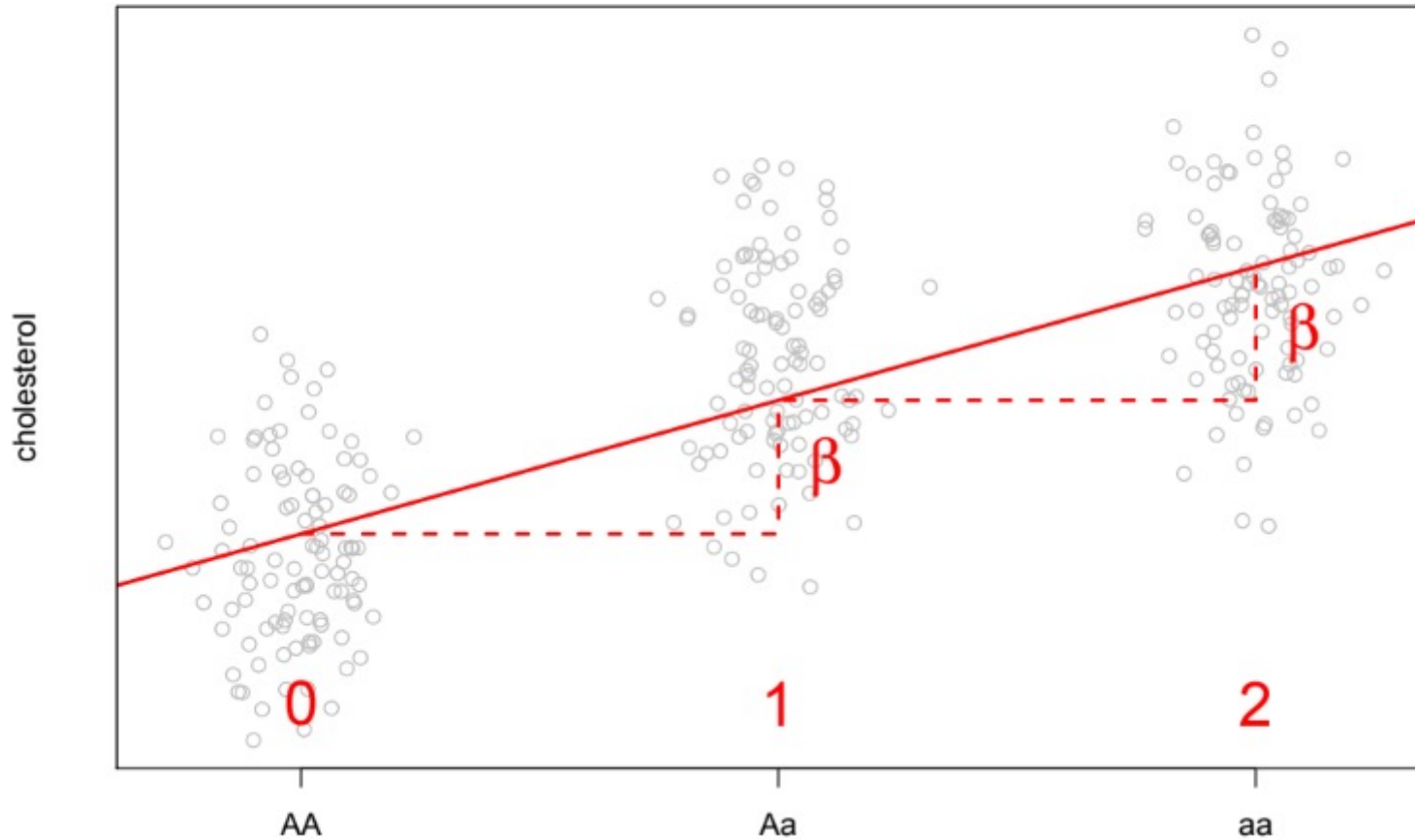
$G_i$ : genotype vector for the  $i$ th individual

- **Continuous Traits**

- Height, blood pressure, BMI, cholesterol, etc
- Like standard linear regression
  - Estimates the  $\beta$
  - Include a single genetic marker (e.g., a variant) as predictor in the model plus other relevant covariates (sex, PCs for ancestry)
  - May want to transform highly skewed variables as this is still linear regression!

# Additive Model

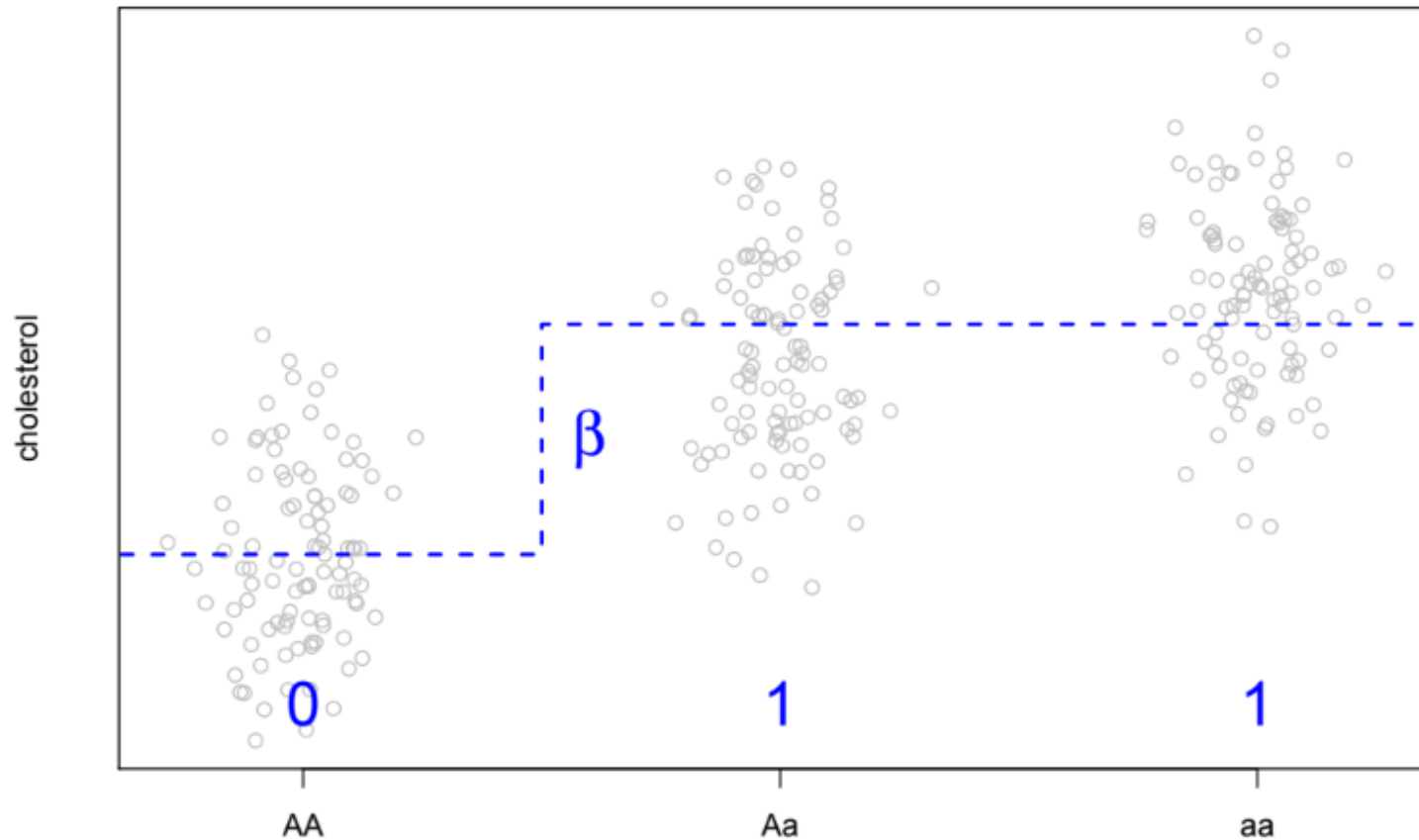
$$y = \beta_0 + \beta \times \# \text{ effect alleles}$$



Regress outcome on # of effect alleles

# Dominant Model

$$y = \beta_0 + \beta \times (G \neq AA)$$



Recessive model counts only the non-effect allele

# In general: we count alleles additively

- The additive model provides the best balance of statistical power, simplicity, and biological relevance for complex traits → remains the most reliable baseline and practical compromise
- Even when underlying molecular mechanisms involve complex gene interactions, the cumulative phenotypic variance is largely captured by an additive approximation
- This is the default in many programs
  - Some have options to specify dominant or recessive models

# Logistic Regression

- **Binary Traits**
  - Disease status, red hair color, morning person, etc
- Estimates an odds ratio (OR), describing how the odds of the event are multiplied per additional copy of the effect allele (the additive model), averaged over the population
- For example,
  - OR of the T allele = 1.20: each additional copy of T multiplies the odds of the event by 1.20, so a person with one T allele has 20% higher odds than a person with zero T alleles, and a person with two T alleles has  $1.20 \times 1.20 \approx 44\%$  higher odds than someone with zero

# Survival Analysis

- “Time-to-event” outcomes
  - Time to death
  - Time to remission
  - Progression-free survival after intervention
- Particularly relevant in cancer survivorship and late-effects research
- Tools:
  - R packages: gwasurvivr, genipe, SurvivalGWAS\_SV
  - Starting with version 4.0, REGENIE supports survival analysis!
- Common pitfalls: convergence issues, restrictive model assumptions (e.g., proportional hazards), and high computational cost at genome-wide scale which grows drastically with sample size

# Mixed-Effect Model

Linear mixed model:

$$Y_i = X_i\alpha + G_i\beta + \mathbf{b}_i + \epsilon_i$$

$b$ : random genetic effect,  $b \sim N(0, \tau \psi)$ ,  $\psi$  is genetic relationship matrix (GRM)

- Widely used to analyze correlated data with multiple sources of variance; in GWAS, that correlation comes from relatedness and shared ancestry between individuals
- How it improves GWAS accuracy: a **Genetic Relationship Matrix (GRM)**, estimated from genome-wide data, captures the degree of genetic similarity between every pair of individuals. Including this as a random effect models relatedness and fine-scale population structure directly
  - Implemented for both quantitative traits (LMMs) and binary traits (GLMMs) across many GWAS programs
  - Pose computational challenges including substantial time complexity and high memory consumption, especially as sample size grows

Stay tuned for a Whole-Genome Regression (WGR) framework at our practicum!

GWAS  
Programs:  
this is a  
**continually  
evolving area**

PLINK (2007)

SNPTEST (2007)

GENESIS (2015)

BOLT-LMM (2015)

SAIGE (2018)

FastGWA-GLMM (2019)

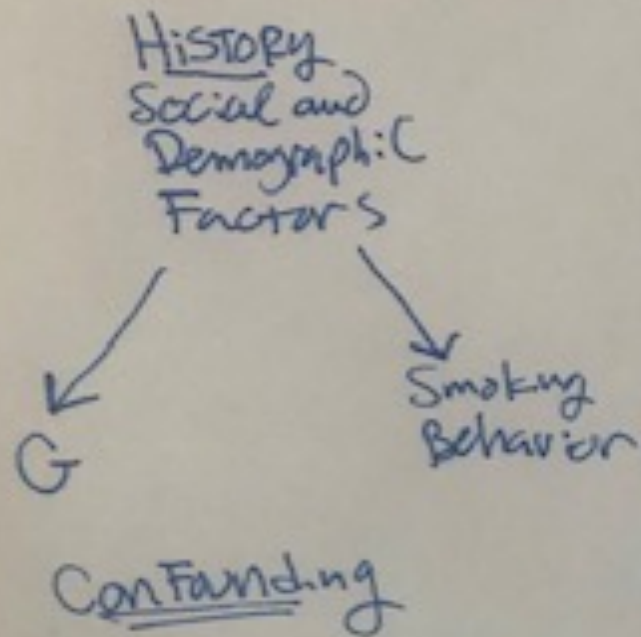
REGENIE (2021) → Used in today's practicum!

# Covariates

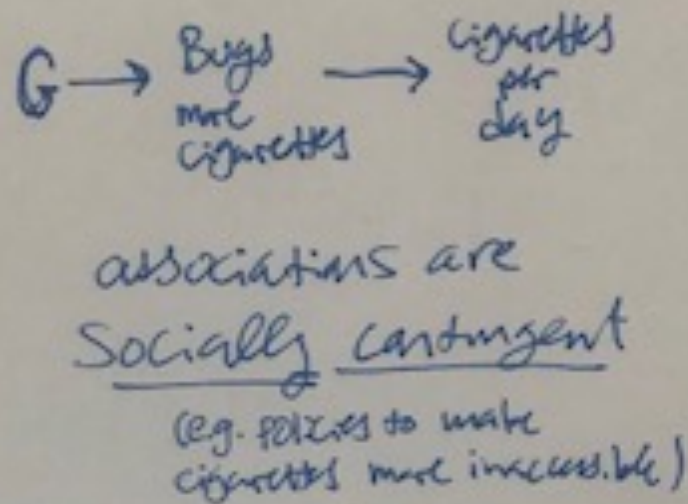
# What goes into the model?

- Consider using directed acyclic graphs (DAGs)
- Do not throw all the variables available into the model as covariates

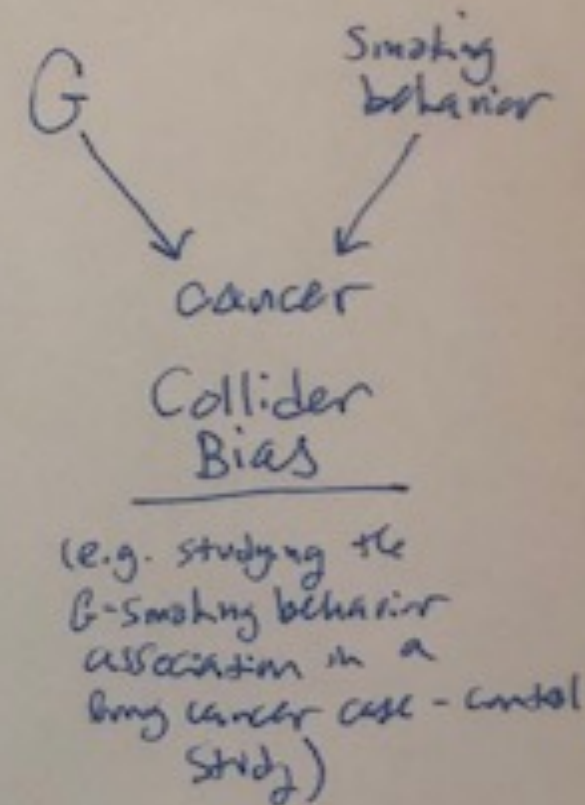




Confounding



Associations are  
Socially Contingent



Collider Bias

Adjusting for “heritable  
covariates” can also induce  
non-causal associations

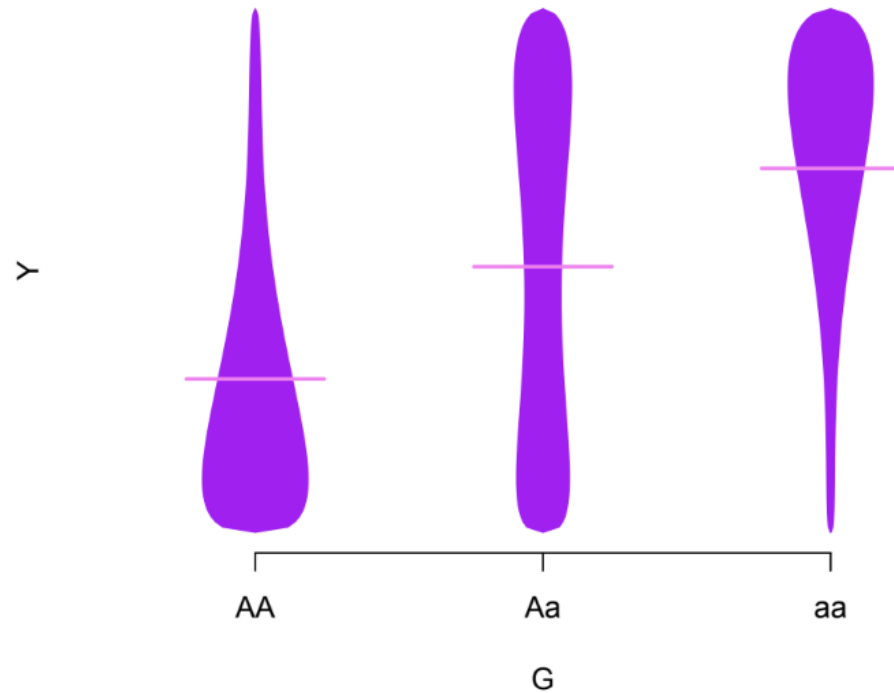
Kuo & Feingold (2010) Genet Epidemiol; Pirinen (2012) Nat Genet;  
Aschard (2015) Am J Hum Genet; Monsees (2009) Genet Epidemiol

# A framework for selecting GWAS covariates

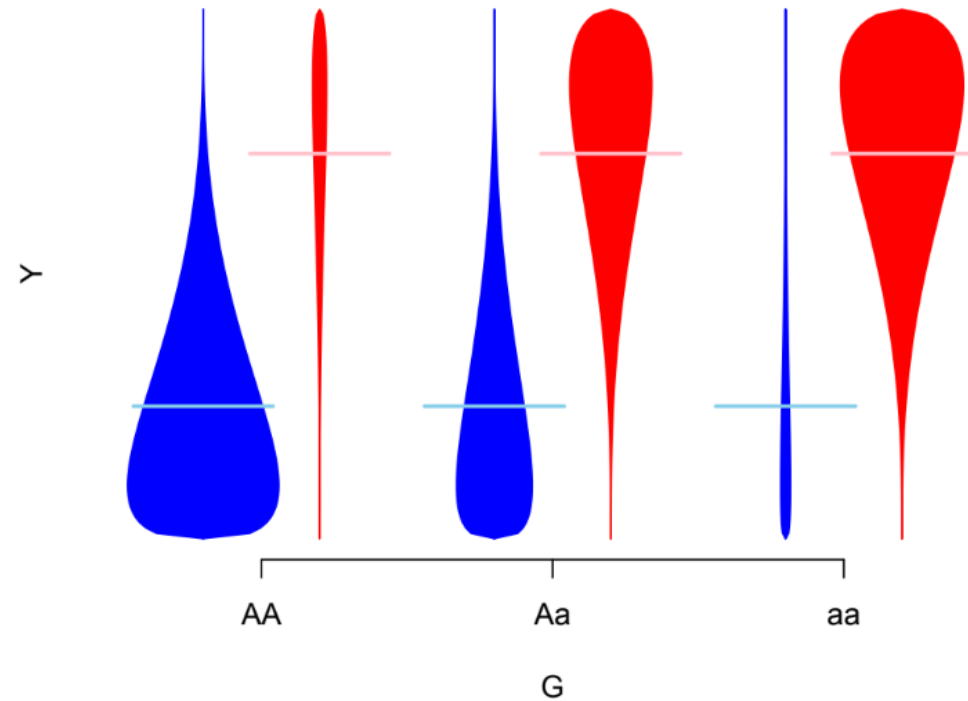
- **Adjust for** covariates that act as common causes of both the genotype and phenotype (e.g., population stratification via genetic PCs)
- **Do not adjust for** covariates on the direct causal pathway between the genotype and phenotype, as this will attenuate or completely eliminate your association signal
- **Do not adjust for** covariates that are common effects of both the genotype and phenotype, as conditioning on them induces collider bias
- **Consider adjusting for** covariates that improve statistical efficiency
  - Age and sex as examples: they often explain substantial variance in the phenotype, and adjusting for them increases statistical power by reducing residual noise/variance
  - Particularly relevant to continuous traits

# Population stratification confounds results in GWAS

Population with a **true association** between phenotype (Y) and genotype (G)



Population with a **confounded association** between phenotype (Y) and genotype (G) driven by ancestry



How can we test for population stratification in our GWAS results?

# Claiming Statistical Significance

# Multiple Testing Correction for Type I Error Control

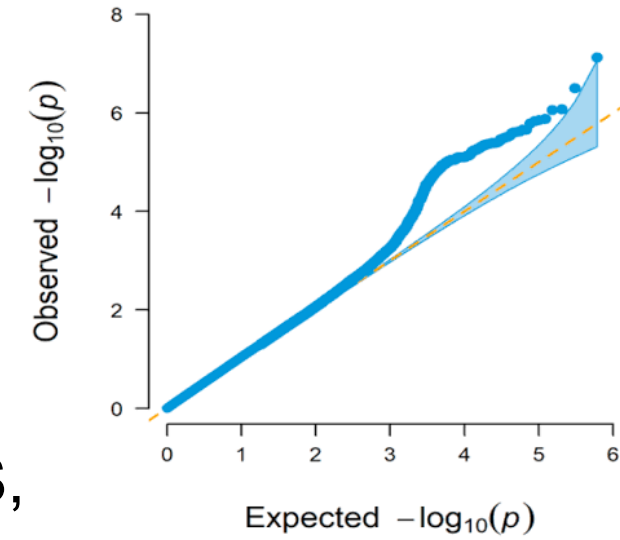
- Different statistical procedures accounting for multiple testing have been used in the genome-wide setting:
  - Bonferroni correction
  - False discovery rate (FDR) procedures
  - Permutation based-approaches
  - Bayesian approaches

# Multiple Testing Correction: Why $5 \times 10^{-8}$ ?

- Bonferroni correction is overly conservative due to LD
- Statistical modeling in early European cohorts estimated roughly 1,000,000 independent testing blocks across the human genome ( $0.05 / 1,000,000 = 5 \times 10^{-8}$ )
- Limitations of the standard threshold:
  - Ancestry-specific variation
    - LD patterns differ across ancestry
  - Allele frequency variation
    - Common variants are much more likely to be highly correlated, inclusion of rarer markers may lead to false positives

# GWAS Results Visualization

# Quantile-Quantile plots (Q-Q plots)



- Essential diagnostic tool for detecting problems in a GWAS, such as population stratification or genotyping artifacts
- In GWAS, Q-Q plots compare the sorted **observed** distribution of test-statistic p-values (typically plotted as  $-\log_{10}(p)$  on the y-axis) **against the** distribution **expected** under the null (on the x-axis)
- Under the null hypothesis of no association, p-values follow a uniform distribution between 0 and 1, i.e., any value is equally likely
  - In GWAS, the vast majority of markers are expected **not** to be associated with the trait, so most points should fall along the diagonal and only the extreme upper tail should deviate above the line

# Lambda ( $\lambda$ ): Related to the Q-Q plot

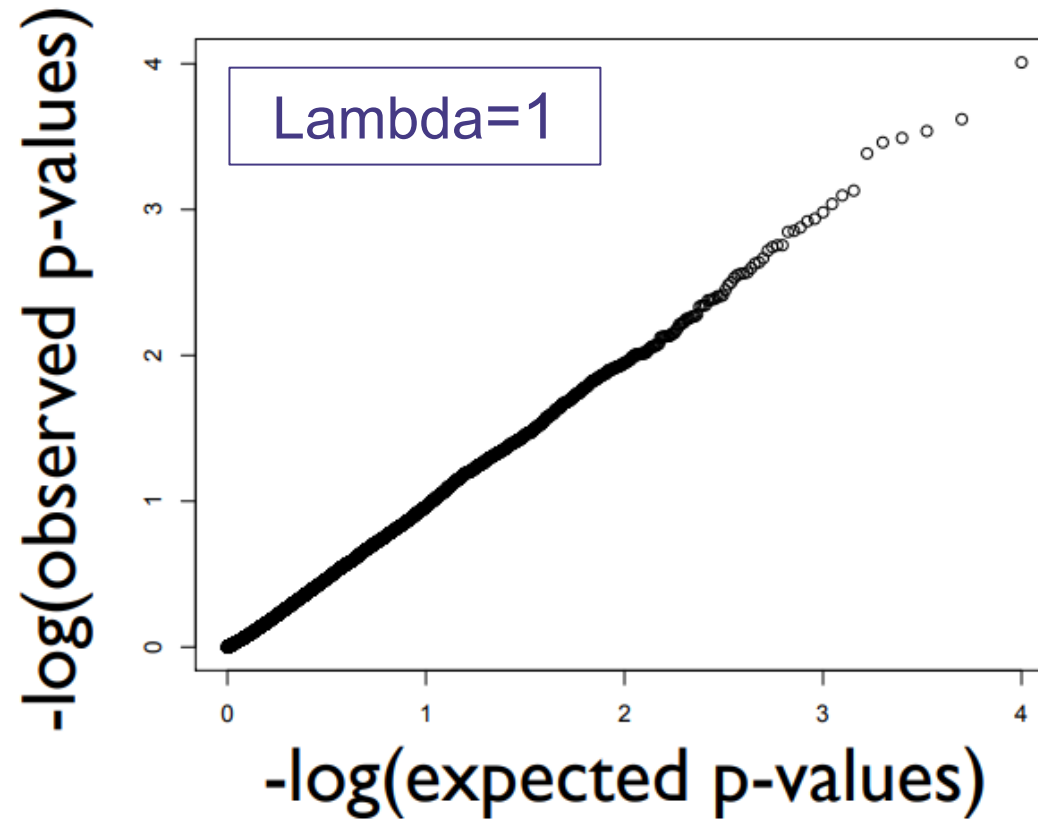
- $\lambda_{GC}$  = the “genomic control inflation factor” = median (observed  $\chi^2$  statistic) / median (expected  $\chi^2$  under the null) = a single-number summary of the Q-Q plot’s overall deviation from the diagonal
- $\lambda_{GC}$  should be close to 1
  - **Inflation ( $\lambda > 1.1$ )**
    - Indicates test statistics are systematically more significant than expected under the null
    - Usually reflects uncontrolled population stratification, cryptic relatedness, or unknown family structure
    - Highly polygenic traits can also inflate  $\lambda$  even in a clean study (should be interpreted relative to sample size  $\rightarrow \lambda_{1,000}$ ): more on this when we cover LD Score Regression session #5

# Lambda ( $\lambda$ ): Related to the Q-Q plot

- $\lambda_{GC}$  should be close to 1
  - **Deflation ( $\lambda < 0.9$ )**
    - Indicates test statistics are systematically less significant than expected (the distribution sits below the diagonal)
    - Usually a sign the test is too conservative, e.g., over-correction for population structure (too many PCs, or an over-fit mixed model), very small samples, or low-MAF variants with poor  $\chi^2$  approximation

# Q-Q plots: GWAS with no detectable hits

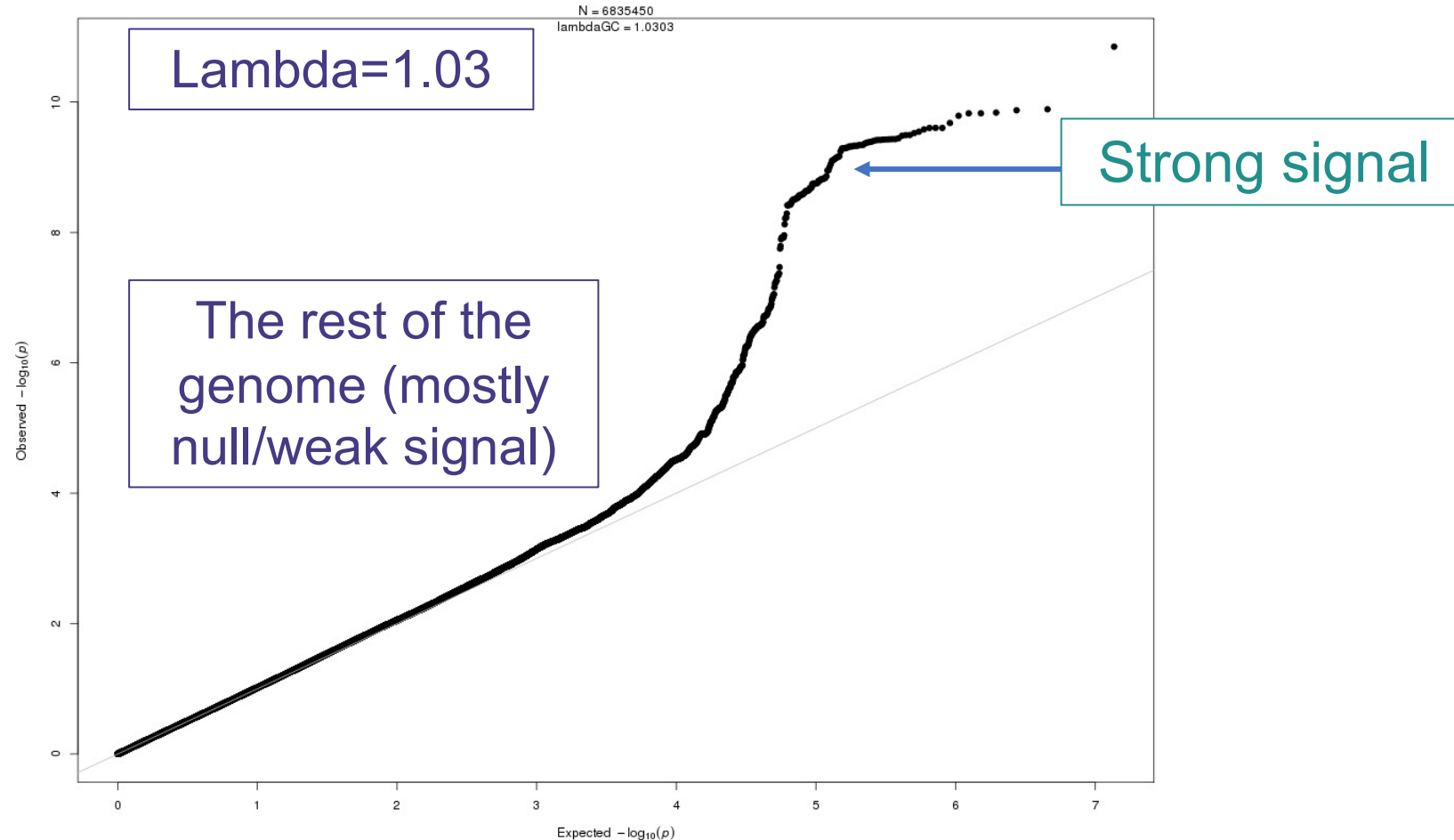
In a correctly performed GWAS with no detectable causal signal, the observed p-values match the null expectation and the Q-Q plot falls right on the diagonal:



Note: This is also what you see when the study is simply underpowered. A clean diagonal means “no problems and no hits,” not necessarily “no biology.”

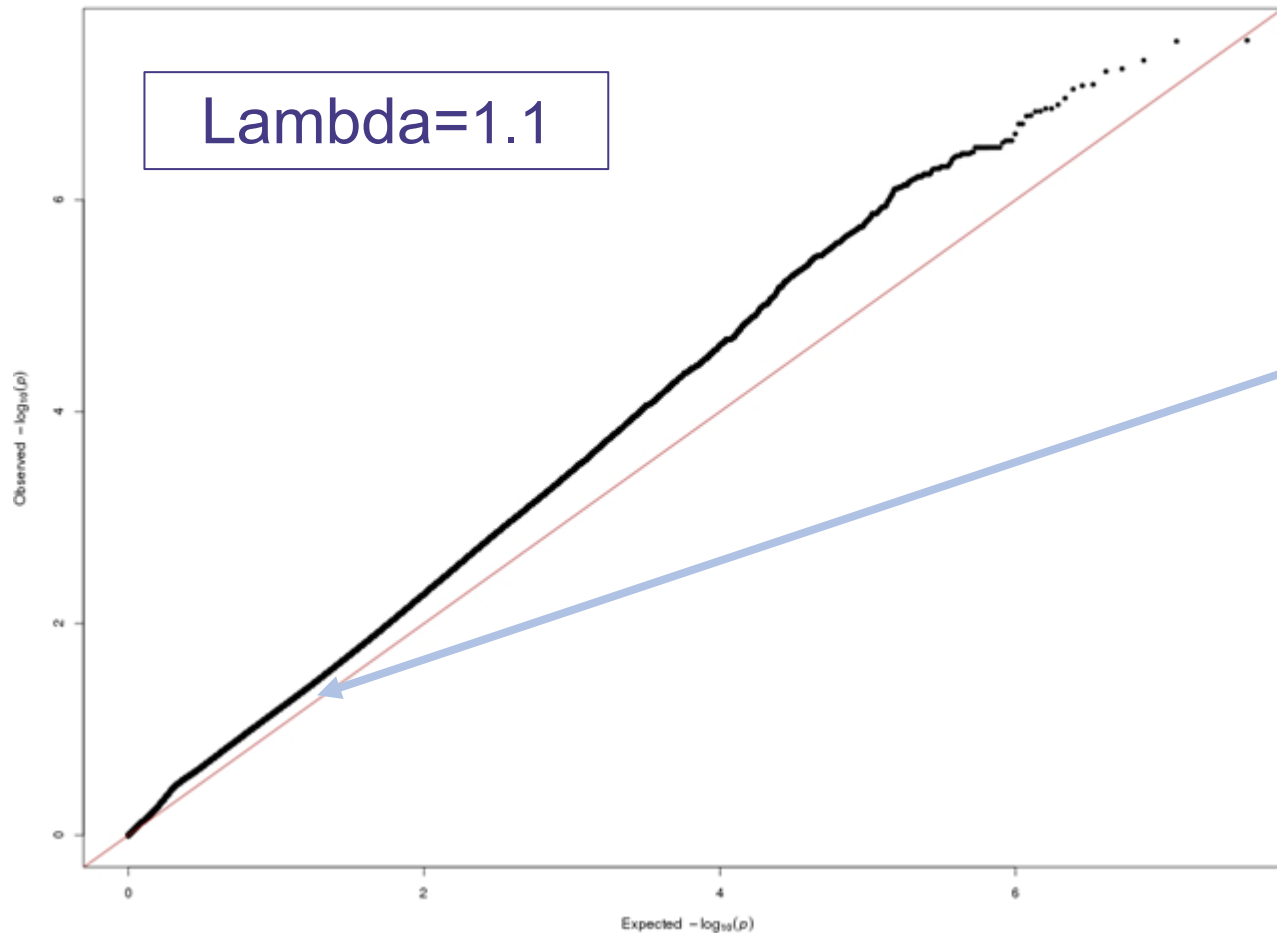
# Q-Q plots: GWAS with significant associations

In a correctly performed GWAS with real causal loci, the Q-Q plot follows the diagonal for most of its length, then lifts off sharply in the upper-right tail:



This is how you want your Q-Q plot to look: on the line, with a clean tail.

# Q-Q plots with inflation: Be wary



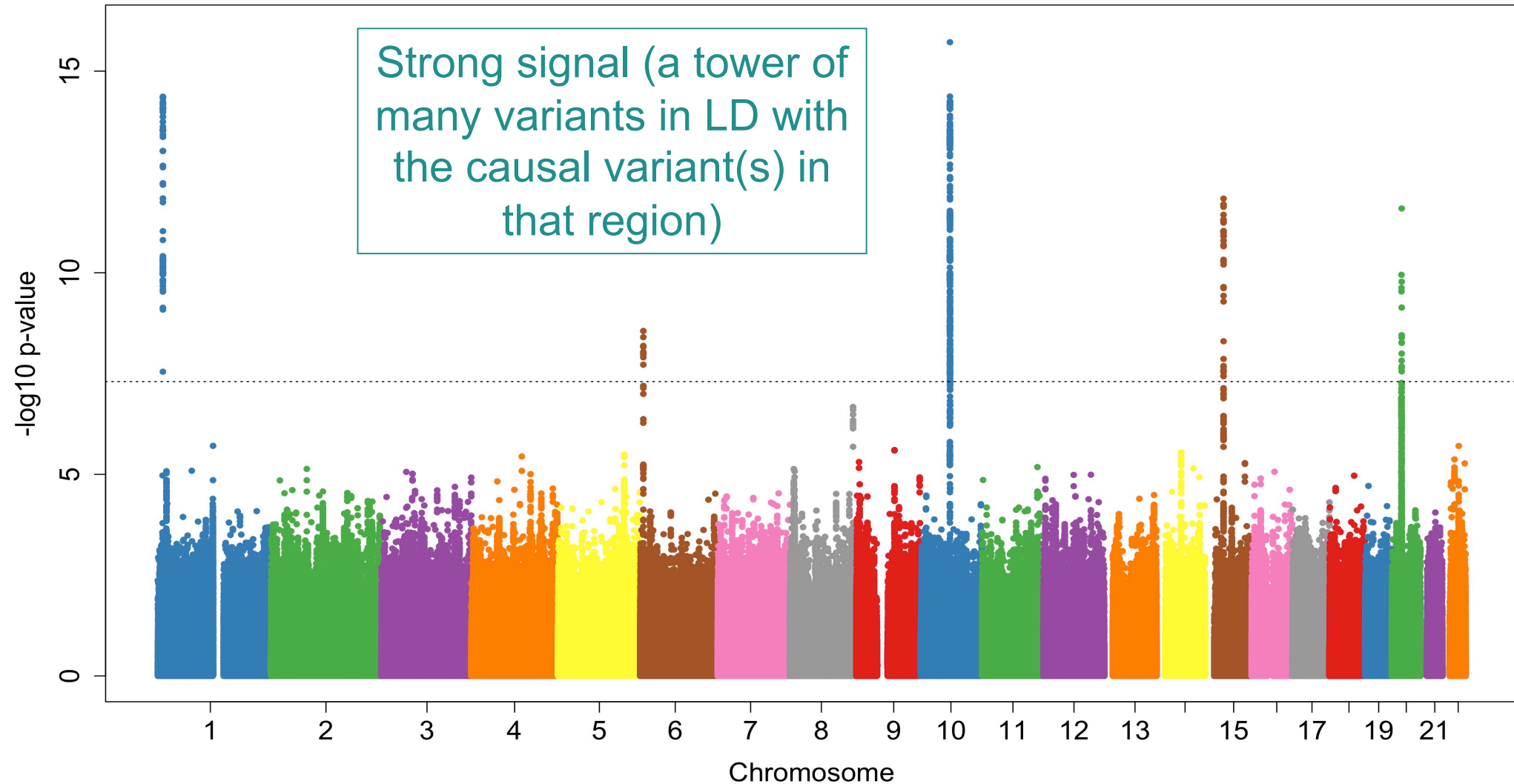
- Inflation!: Early separation of the expected from the observed values means many moderately significant  $p$ -values are more significant than the null predicts
- Rarely due to thousands of true positives; more often due to systematic bias that must be corrected before trusting any hits

# Manhattan Plots

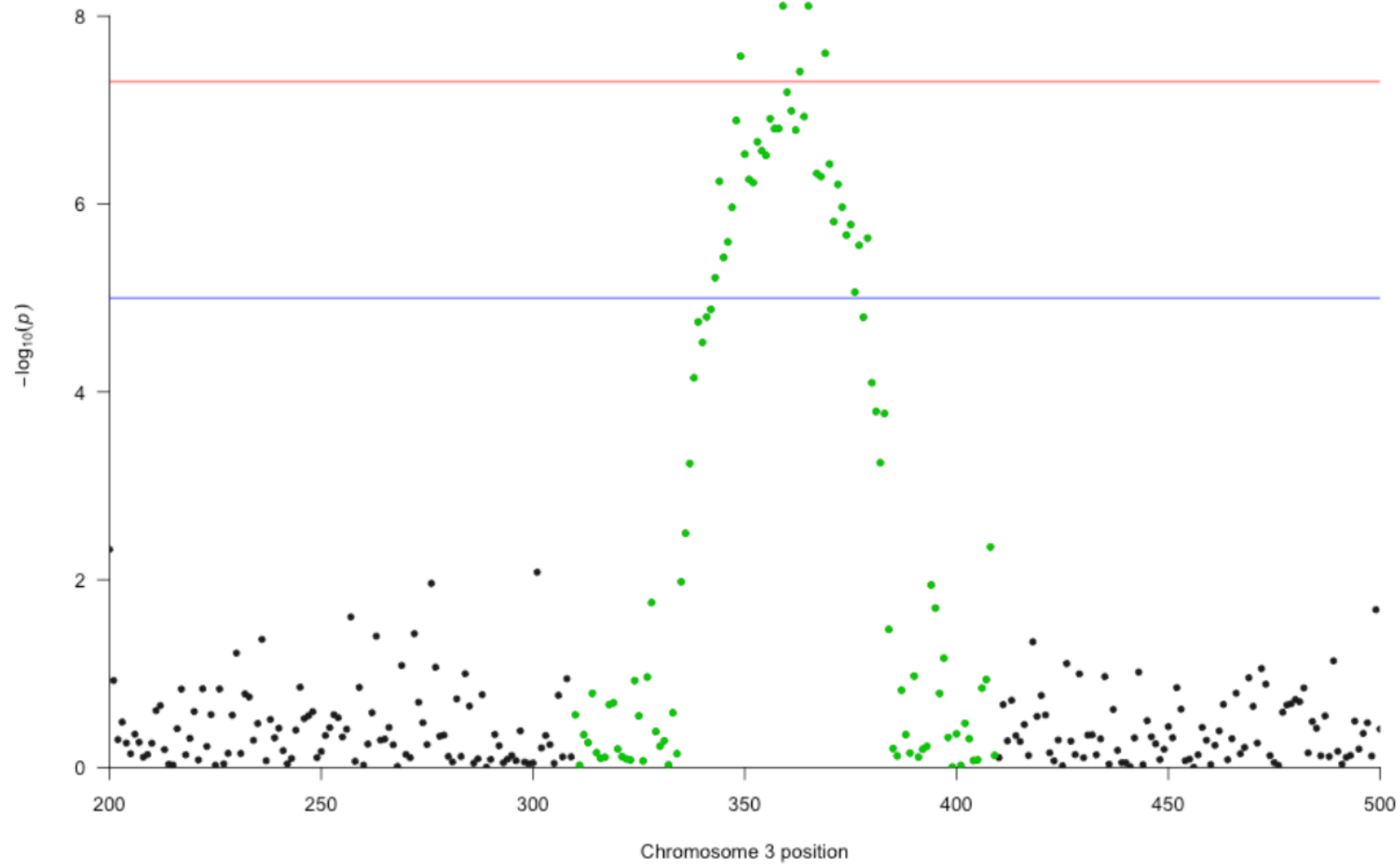


- Plots each variant's genomic position against its association p-value
- Each dot represents a genetic variant, ordered along the x axis by chromosome and base-pair position
- The y axis shows association strength as the  $-\log_{10}(\text{p-value})$ ; smaller p-values plot higher, so the tallest peaks are the strongest signals
- Usually includes a horizontal line marking the genome-wide significance threshold of  $p < 5 \times 10^{-8}$

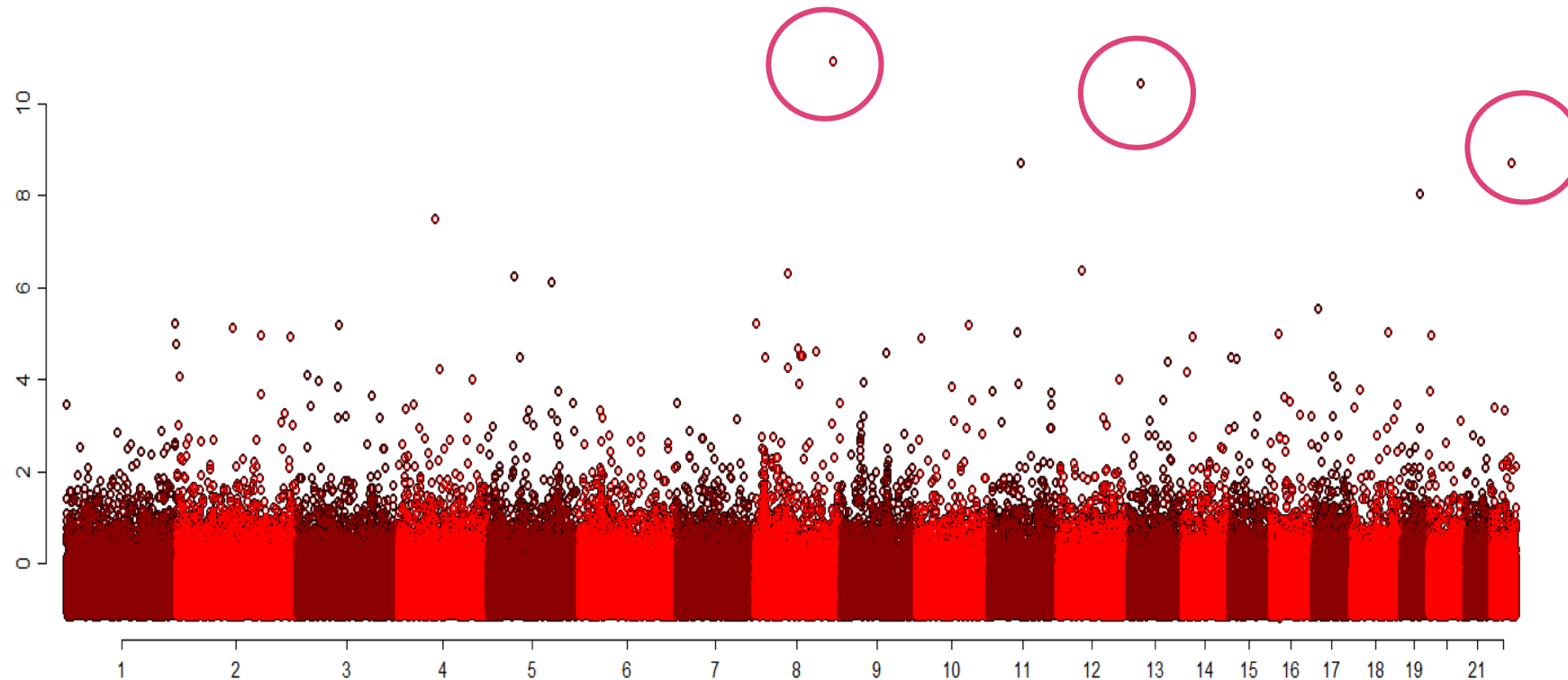
# Manhattan Plots



# Regional Plots: Zoomed-in Manhattan Plot

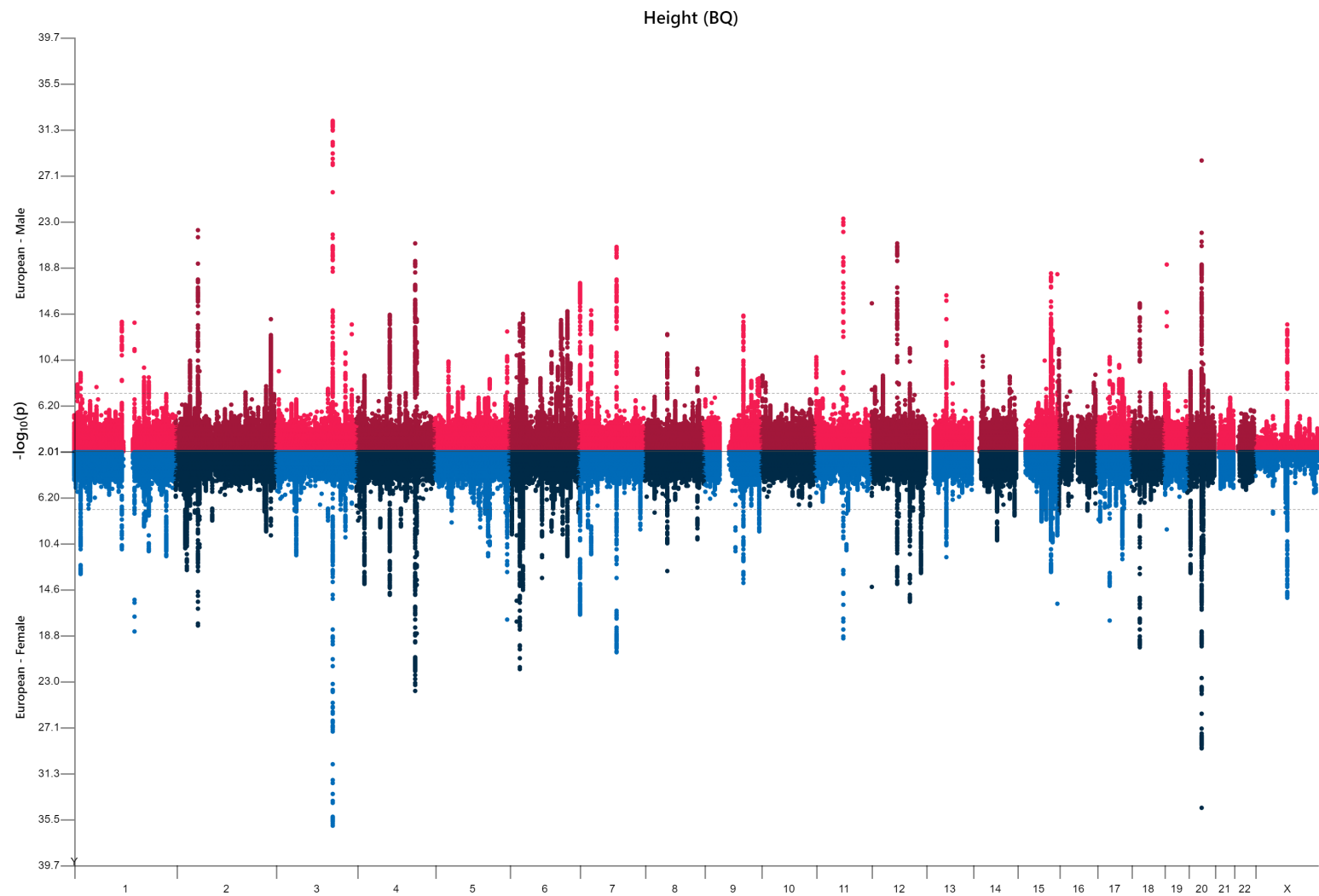


# Manhattan Plots



Remember: There are no stars over Manhattan!

# Miami Plot: Adult Height Male vs. Female



# Meta-Analysis

# Why meta-analysis?

- **Sample size** drives discovery
  - Complex-trait effects are tiny, most single cohorts aren't large enough to find most susceptibility loci
  - Consortia and meta-analyses have driven most locus discovery
- **Privacy/consent/legal limits** often block sharing individual-level data
  - Federated / summary-level analysis lets studies contribute per-variant results instead of raw genotypes

# Meta-analysis vs. mega-analysis

- Two ways to combine studies:
  - Mega-analysis: pool individual-level data; statistically ideal but often impractical (privacy, harmonization, logistics)
  - Meta-analysis: combine summary statistics; highly feasible, with little loss of information vs. mega-analysis
- The catch at scale:
  - Biases negligible in small studies become detectable and misleading at very large N → minor uncorrected confounding or artifacts get amplified into apparent signal
  - Calibration checks ( $\lambda$ , Q-Q) and QC matter more, not less, as samples grow

Reviews of GWAS consortia: MA Austin (2012) *Am J Epidemiol*; SN Bennett (2011) *Genet Epidemiol*; E Ingleson (2011) *Circulation: Cardiovascular Genetics*; PF Sullivan (2010) *Neuron*.  
On the relative merits of mega- versus meta-analysis: DY Lin (2009) *Genet Epidemiol*

# Imputation: what makes it possible

- Imputation is what lets studies on different arrays be combined
- How it works: individuals share **haplotype stretches** from common ancestors; because nearby alleles are correlated (LD), algorithms use a dense reference panel (e.g., 1000 Genomes, TOPMed) to infer genotypes a study didn't directly measure
- Why it matters: brings cohorts genotyped on different chips onto a common variant set, so their summary statistics become combinable

# Imputation for cross-chip comparisons, e.g. for meta-analysis of SNP associations

All 1KGP SNPs



Illumina SNPs



Affymetrix SNPs



Overlap SNPs



# Imputation for cross-chip comparisons, e.g. for meta-analysis of SNP associations

All 1KGP SNPs



Illumina SNPs



Affymetrix SNPs



Overlap SNPs



# Imputation outputs & uses

**Input:** 2,000 genotyped samples, 500K (0.5M) SNPs

**Reference panel:** TOPMed r3 (133,597 reference samples, ~446M variants)

→ **Output:** same 2,000 samples, now with genotypes at up to 446M variants

Output format (per unobserved genotype):

- Dosage (continuous, 0–2): expected allele count, e.g., if a variant has a 60% chance of being 1 and a 40% chance of being 2, the dosage is 1.4
- Hard call (discrete, 0–2): the most likely genotype

**How to analyze imputed data appropriately?**

- Regress outcome on “dosage” to propagate imputation uncertainty directly into the regression and correctly inflate SE for poorly-imputed variants rather than false confidence

# Imputation quality & its role in meta-analysis

## Are all imputed genotypes equally reliable?

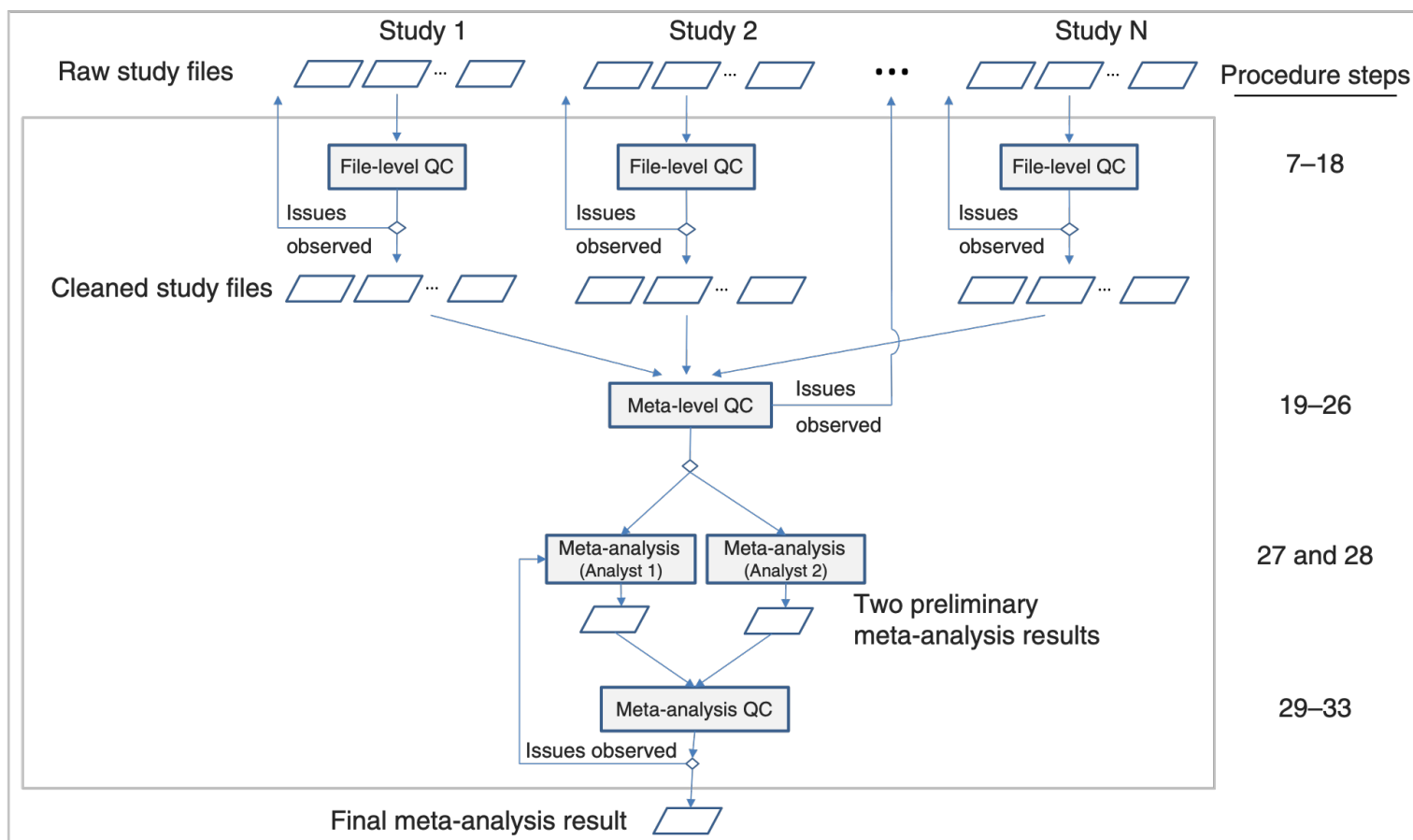
- **No.**  $R^2$  estimates imputation quality as the squared correlation between imputed dosage and the true (unobserved) genotype. Low  $R^2$  means the imputation shouldn't be trusted at that variant.

## Why this matters for meta-analysis?

- Imputation **accuracy drops for** low-MAF variants and for samples that are poorly matched to the reference panel's ancestry.
- When two cohorts being meta-analyzed differ in imputation accuracy **at the same variant**, it can look like a real between-study effect, or even confound with case/control status if cases and controls were imputed differently → **filter** on  $R^2$  (and MAF) consistently across cohorts before combining studies

# Quality control and conduct of genome-wide association meta-analyses

Thomas W Winkler<sup>1</sup>, Felix R Day<sup>2</sup>, Damien C Croteau-Chonka<sup>3,4</sup>, Andrew R Wood<sup>5</sup>, Adam E Locke<sup>6</sup>, Reedik Mägi<sup>7</sup>, Teresa Ferreira<sup>8</sup>, Tove Fall<sup>9,10</sup>, Mariaelisa Graff<sup>11</sup>, Anne E Justice<sup>11</sup>, Jian'an Luan<sup>2</sup>, Stefan Gustafsson<sup>9</sup>, Joshua C Randall<sup>12</sup>, Sailaja Vedantam<sup>13–15</sup>, Tsegaselassie Workalemahu<sup>16</sup>, Tuomas O Kilpeläinen<sup>17</sup>, André Scherag<sup>18,19</sup>, Tonu Esko<sup>7,13–15</sup>, Zoltán Kutalik<sup>20–22</sup>, Iris M Heid<sup>1,27</sup>, Ruth J F Loos<sup>23–25,27</sup> & the Genetic Investigation of Anthropometric Traits (GIANT) Consortium<sup>26</sup>



**Software Tools**  
(handle the tedious bits like making sure ref/non-ref alleles are the same across different studies)

[PLINK](#)  
[EASYQC](#)  
[METAL](#)  
[METASOFT](#)

# Meta-analysis: Fixed Effects

- **Fixed effects** assumes there is **one true effect size**, shared by every cohort (cohorts differ only in how precisely they estimate it)
  - Combining cohorts by weighting each one's contribution by its precision (larger, less noisy studies count more) gives a more powerful test and a more precise estimate, provided the true effect is genuinely shared
  - Two common ways to weight:
    - Inverse-variance weighting (**IVW**): the standard approach, weights by sample size/precision
    - Weighted Z-scores (**Stouffer's method**): a simpler alternative used when studies measured things on different scales

# Meta-analysis of GWAS Results

“Fixed effects” meta-analysis has been the default in GWAS to maximize power, even when signals are somewhat heterogeneous across studies.

$$z_{\text{meta}} = \frac{\langle \beta \rangle}{\langle \text{SE} \rangle}$$
$$\langle \beta \rangle = \frac{\sum_i [\beta_i / (\text{SE}_i)^2]}{\sum_i [1 / (\text{SE}_i)^2]}$$
$$\langle \text{SE} \rangle = \sqrt{\frac{1}{\sum_i [1 / (\text{SE}_i)^2]}}$$

**Weighted beta/SE approach:** combines each study’s effect estimate, weighted by precision. Appropriate here as the goal is testing whether there’s any true association (the “global null”), not modeling study-to-study variation in the effect

$$z_{\text{meta}} = \sum_i z_i \times w_i$$
$$w_i = \sqrt{\frac{N_i}{N_{\text{total}}}}$$

**Weighted Z scores (“Stouffer’s method”):** an alternative when studies used different measurement scales (e.g., Olink’s relative NPX values and ELISA absolute concentrations for the same protein)

De Bakker (2008) *Hum Mol Genet*  
Non-standard random effects model for testing global null:  
Han and Eskin (2011) *Am J Hum Genet*

# Meta-analysis: Random Effects

- **Random effects** assumes the **true effect size itself varies** somewhat across cohorts (e.g., due to differences in ancestry or environment)
- **Older method** (DerSimonian & Laird):
  - **H0**: The average effect across all possible studies is zero
  - This method allows for the true effects of individual studies to be completely different from one another ( $\tau^2 > 0$ ), but tests whether their overall average centers exactly at zero
  - When studies disagree, the DL method calculates a large between-study variance ( $\tau^2$ ) and adds it to the SE of every single study. This means that even if a true effect exists, the DL method treats the conflict between studies as “noise”
- **overly conservative** by assuming studies disagree even under the null of no effect

# Meta-analysis: Random Effects

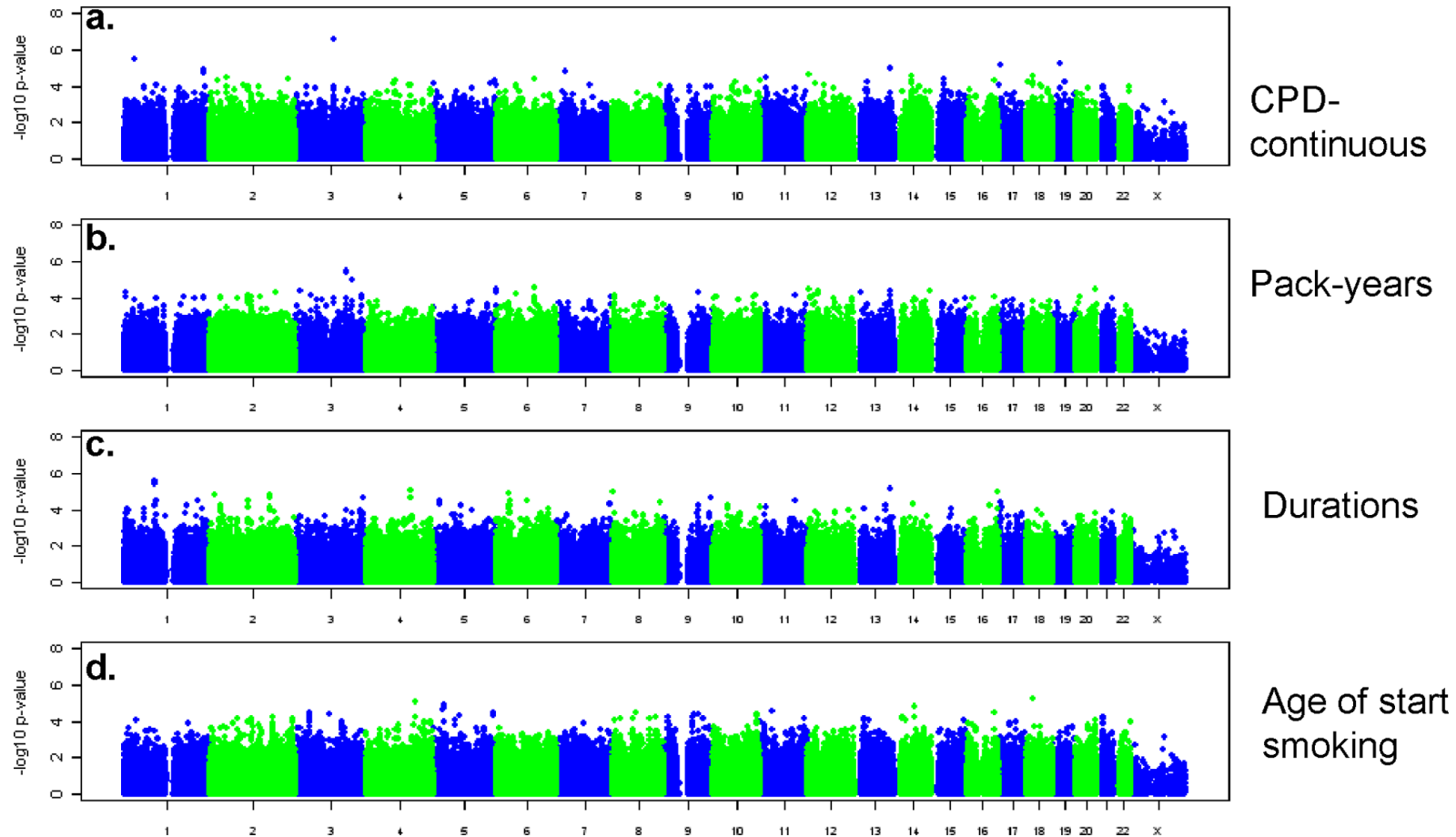
- **Newer method** (Han & Eskin, in METASOFT): → Used in today's practicum!
  - **RE2** (Random Effects 2):
    - **H0**: No effect in any study AND no variation ( $\tau^2 = 0$ ) between cohorts
    - Uses a two-degree-of-freedom joint test to simultaneously flag an overall effect and genuine heterogeneity
  - **BE** (binary effect):
    - **H0**: The effect is exactly zero in ALL cohorts
    - Key metric (m-value): the posterior probability (0 to 1) that the genetic effect actually exists in a specific study
- **Stricter Null Hypotheses**: Instead of assuming study disagreement as a baseline, both methods test against a rigid null of absolute zero
- **Boosted Statistical Power**: Drastically outperforms Fixed Effects and traditional Random Effects when true genetic effects vary across cohorts

# Examples of increasing N via meta-analysis

Two case studies showing how successive meta-analyses drove discovery by scaling up N:

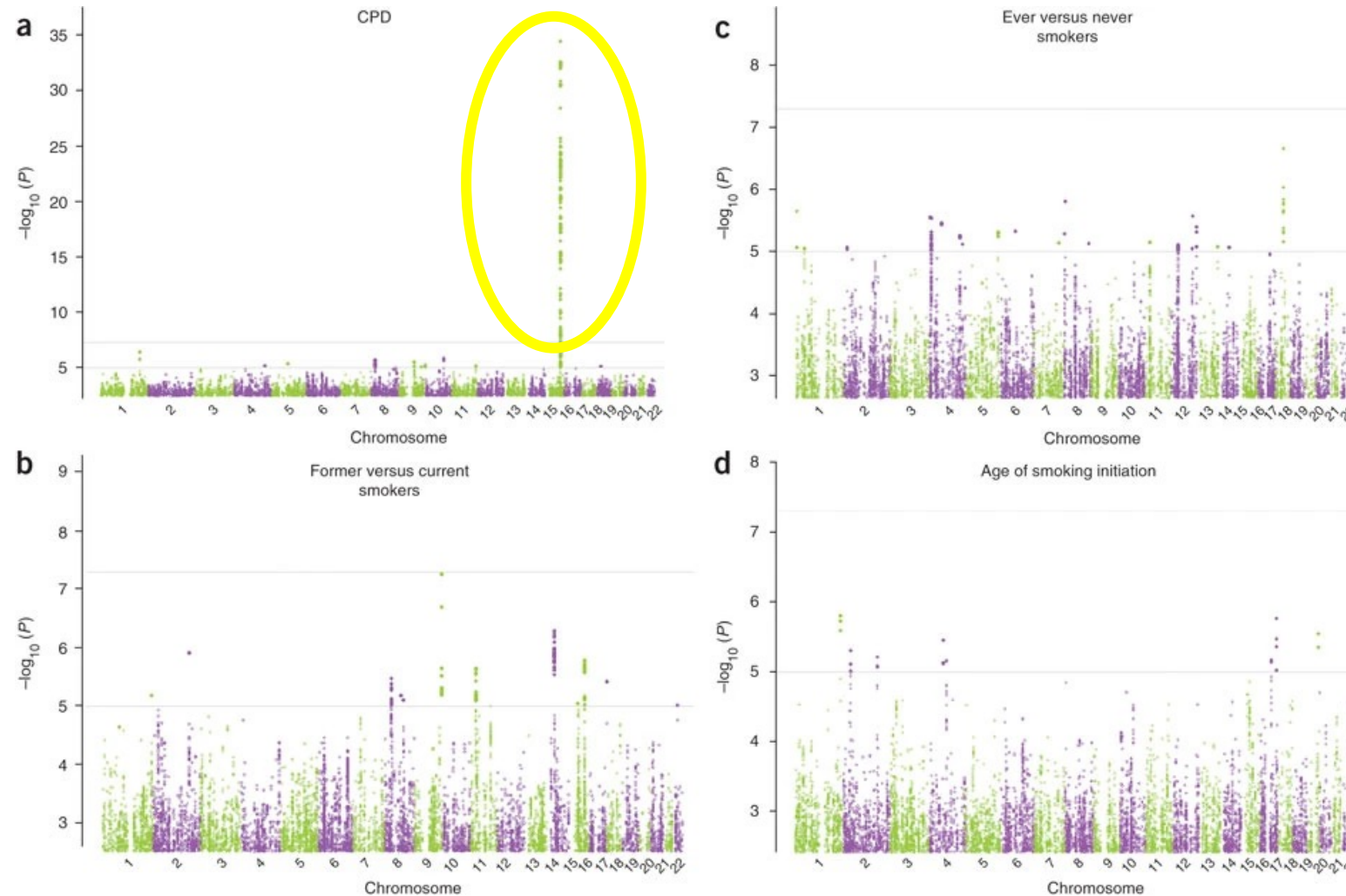
1. **Successively larger GWAS of smoking behavior:** small study → consortium → multi-million-person meta-analysis
2. **A saturated map of common genetic variants associated with human height:** the GIANT/23andMe endpoint of that same scaling logic

# Smoking GWAS: early iterations



An early smoking-behavior GWAS combining **2,300 males** (PLCO) and **2,300 females** (NHS) as a case-control meta-analysis, showing **no genome-wide-significant hits**

# Meta-analysis of smoking GWAS



The TAG Consortium meta-analyzed **74,503 subjects across 16 studies** → a clear, sharp peak emerges at *CHRNA3-CHRNA4-CHRNA5* for cigarettes-per-day

# Meta-analysis of smoking GWAS

**Table 2 Meta-analytic results from three GWAS smoking consortia**

| SNP                                                                | Alleles | TAG meta-analysis |          |         |       |                          | Ox-GSK meta analysis |         |       |                          | ENGAGE meta analysis |         |       |                          | Combined results |         |       |                          |
|--------------------------------------------------------------------|---------|-------------------|----------|---------|-------|--------------------------|----------------------|---------|-------|--------------------------|----------------------|---------|-------|--------------------------|------------------|---------|-------|--------------------------|
|                                                                    |         | Coded AF          | <i>n</i> | β       | s.e.  | <i>P</i> value           | <i>n</i>             | β       | s.e.  | <i>P</i> value           | <i>n</i>             | β       | s.e.  | <i>P</i> value           | <i>n</i>         | β       | s.e.  | <i>P</i> value           |
| CPD <sup>a</sup> : <i>CHRNA3</i>                                   |         |                   |          |         |       |                          |                      |         |       |                          |                      |         |       |                          |                  |         |       |                          |
| rs1051730                                                          | G/A     | 0.65              | 38,181   | −1.0207 | 0.086 | 8.00 × 10 <sup>−33</sup> | 14,952               | −1.1593 | 0.139 | 8.88 × 10 <sup>−17</sup> | 20,720               | −0.9648 | 0.089 | 2.15 × 10 <sup>−27</sup> | 73,853           | −1.0209 | 0.056 | 2.75 × 10 <sup>−73</sup> |
| rs16969968                                                         | G/A     | 0.65              | 38,181   | −1.0150 | 0.085 | 4.48 × 10 <sup>−33</sup> | 14,952               | −1.1153 | 0.137 | 3.72 × 10 <sup>−16</sup> | 20,720               | −0.9426 | 0.089 | 2.07 × 10 <sup>−26</sup> | 73,853           | −1.0029 | 0.056 | 5.57 × 10 <sup>−72</sup> |
| CPD <sup>a</sup> : in <i>LOC100188947</i>                          |         |                   |          |         |       |                          |                      |         |       |                          |                      |         |       |                          |                  |         |       |                          |
| rs1329650                                                          | T/G     | 0.28              | 38,181   | −0.4317 | 0.091 | 2.33 × 10 <sup>−6</sup>  | 14,952               | −0.2568 | 0.145 | 7.61 × 10 <sup>−2</sup>  | 20,720               | −0.3464 | 0.092 | 1.73 × 10 <sup>−4</sup>  | 73,853           | −0.3673 | 0.059 | 5.67 × 10 <sup>−10</sup> |
| rs1028936                                                          | C/A     | 0.18              | 37,284   | −0.5545 | 0.116 | 1.57 × 10 <sup>−6</sup>  | 14,952               | −0.2451 | 0.176 | 1.65 × 10 <sup>−1</sup>  | 20,720               | −0.4252 | 0.113 | 1.77 × 10 <sup>−4</sup>  | 72,956           | −0.4464 | 0.074 | 1.29 × 10 <sup>−9</sup>  |
| CPD <sup>a</sup> : <i>EGLN2</i> , near <i>CYP2A6</i>               |         |                   |          |         |       |                          |                      |         |       |                          |                      |         |       |                          |                  |         |       |                          |
| rs3733829                                                          | G/A     | 0.36              | 38,181   | 0.3538  | 0.090 | 7.67 × 10 <sup>−5</sup>  | 14,952               | 0.0477  | 0.145 | 7.43 × 10 <sup>−1</sup>  | 20,720               | 0.4204  | 0.089 | 2.90 × 10 <sup>−6</sup>  | 73,853           | 0.3328  | 0.058 | 1.04 × 10 <sup>−8</sup>  |
| Smoking initiation (ever versus never smokers): <i>BDNF</i>        |         |                   |          |         |       |                          |                      |         |       |                          |                      |         |       |                          |                  |         |       |                          |
| rs6265                                                             | T/C     | 0.21              | 74,035   | −0.0630 | 0.015 | 1.72 × 10 <sup>−5</sup>  | 34,226               | −0.0448 | 0.022 | 4.48 × 10 <sup>−2</sup>  | 34,762               | −0.0762 | 0.024 | 1.39 × 10 <sup>−3</sup>  | 143,023          | −0.0614 | 0.011 | 1.84 × 10 <sup>−8</sup>  |
| rs1013442                                                          | T/A     | 0.26              | 74,035   | −0.0568 | 0.014 | 3.39 × 10 <sup>−5</sup>  | 34,226               | −0.0386 | 0.021 | 6.36 × 10 <sup>−2</sup>  | 34,762               | −0.0674 | 0.020 | 9.60 × 10 <sup>−4</sup>  | 143,023          | −0.0551 | 0.010 | 3.31 × 10 <sup>−8</sup>  |
| rs4923457                                                          | T/A     | 0.23              | 74,035   | −0.0600 | 0.014 | 2.08 × 10 <sup>−5</sup>  | 34,226               | −0.0421 | 0.022 | 5.05 × 10 <sup>−2</sup>  | 34,762               | −0.0752 | 0.024 | 1.91 × 10 <sup>−3</sup>  | 143,023          | −0.0586 | 0.011 | 3.33 × 10 <sup>−8</sup>  |
| rs4923460                                                          | T/G     | 0.23              | 74,035   | −0.0598 | 0.014 | 2.22 × 10 <sup>−5</sup>  | 34,226               | −0.0427 | 0.022 | 4.81 × 10 <sup>−2</sup>  | 34,762               | −0.0734 | 0.024 | 2.51 × 10 <sup>−3</sup>  | 143,023          | −0.0583 | 0.011 | 4.08 × 10 <sup>−8</sup>  |
| rs4074134                                                          | T/C     | 0.23              | 74,035   | −0.0603 | 0.014 | 1.90 × 10 <sup>−5</sup>  | 34,226               | −0.0421 | 0.022 | 5.08 × 10 <sup>−2</sup>  | 34,762               | −0.0725 | 0.024 | 2.81 × 10 <sup>−3</sup>  | 143,023          | −0.0582 | 0.011 | 4.11 × 10 <sup>−8</sup>  |
| rs1304100                                                          | G/A     | 0.26              | 74,035   | −0.0557 | 0.014 | 4.86 × 10 <sup>−5</sup>  | 34,226               | −0.0460 | 0.021 | 2.62 × 10 <sup>−2</sup>  | 34,762               | −0.0651 | 0.022 | 2.88 × 10 <sup>−3</sup>  | 143,023          | −0.0554 | 0.010 | 4.44 × 10 <sup>−8</sup>  |
| rs6484320                                                          | T/A     | 0.24              | 74,035   | −0.0597 | 0.014 | 2.04 × 10 <sup>−5</sup>  | 34,226               | −0.0387 | 0.021 | 6.78 × 10 <sup>−2</sup>  | 34,762               | −0.0723 | 0.024 | 2.13 × 10 <sup>−3</sup>  | 143,023          | −0.0571 | 0.010 | 4.91 × 10 <sup>−8</sup>  |
| rs879048                                                           | C/A     | 0.23              | 74,035   | −0.0598 | 0.014 | 2.28 × 10 <sup>−5</sup>  | 34,226               | −0.0409 | 0.022 | 5.86 × 10 <sup>−2</sup>  | 34,762               | −0.0728 | 0.024 | 2.41 × 10 <sup>−3</sup>  | 143,023          | −0.0578 | 0.011 | 4.94 × 10 <sup>−8</sup>  |
| Smoking cessation (former versus current smokers): near <i>DBH</i> |         |                   |          |         |       |                          |                      |         |       |                          |                      |         |       |                          |                  |         |       |                          |
| rs3025343                                                          | G/A     | 0.84              | 41,278   | 0.1177  | 0.026 | 5.68 × 10 <sup>−6</sup>  | 23,646               | 0.1295  | 0.041 | 1.76 × 10 <sup>−3</sup>  | NA                   | NA      | NA    | NA                       | 64,924           | 0.1210  | 0.022 | 3.56 × 10 <sup>−8</sup>  |

All SNPs coded to NCBI Build 36/UCSC hg18 forward strand. Coded allele frequency refers to the allele analyzed as the predictor allele; it is not necessarily the minor allele. For CPD, 174 SNPs followed up across three consortia; 130 exceeded genome-wide significance and the two top SNPs are presented. NA, not available.

<sup>a</sup>CPD was analyzed as a continuous variable representing the number of cigarettes smoked per day. Smoking initiation and smoking cessation were analyzed as dichotomous variables, contrasting ever versus never and former versus current smokers, respectively.

Combining **three** separate smoking **consortia** (TAG, Ox-GSK, ENGAGE) identified **four additional loci**

**Note:** only meta-analyzed “top” 15 SNPs from each study—a missed opportunity! SNPs that are not near the “top” in any individual study can become significant in meta-analysis

# Meta-analysis of smoking GWAS



## Association studies of up to 1.2 million individuals yield new insights into the genetic etiology of tobacco and alcohol use

Tobacco and alcohol use are leading causes of mortality that influence risk for many complex diseases and disorders<sup>1</sup>. They are heritable<sup>2,3</sup> and etiologically related<sup>4,5</sup> behaviors that have been resistant to gene discovery efforts<sup>6-11</sup>. In sample sizes up to 1.2 million individuals, we discovered 566 genetic variants in 406 loci associated with multiple stages of tobacco use (initiation, cessation, and heaviness) as well as alcohol use, with 150 loci evidencing pleiotropic association. Smoking phenotypes were positively genetically correlated with many health conditions, whereas alcohol use was negatively correlated with these conditions, such that increased genetic risk for alcohol use is associated with lower disease risk. We report evidence for the involvement of many systems in tobacco and alcohol use, including genes involved in nicotinic, dopaminergic, and glutamatergic neurotransmission. The results provide a solid starting point to evaluate the effects of these loci in model organisms and more precise substance use measures.

GSCAN consortium  
(**>30 studies**, ~1.2M  
individuals), identifying  
**406 loci**

# A saturated map of common genetic variants associated with human height

## Meta-analysis identifies 12,111 height-associated SNPs

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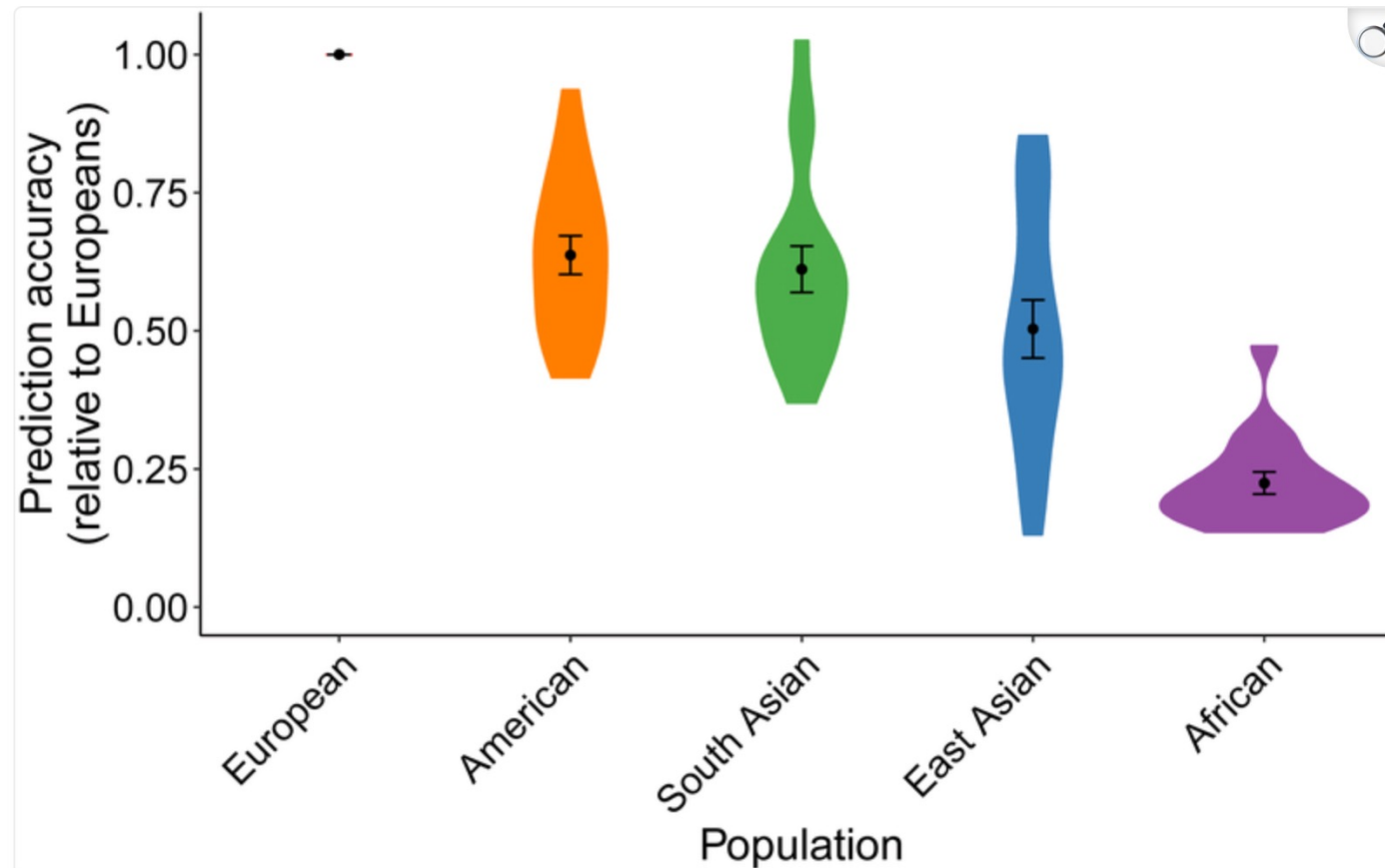
We performed genetic analysis of up to 5,380,080 individuals from 281 studies from the GIANT consortium and 23andMe.

million individuals from multiple ancestries.<sup>5</sup> This unprecedentedly large study showed a saturation of the common-variant architecture among European-ancestry genomes, whereby approximately 12,000 SNPs jointly explain 40% of variation in out-of-sample prediction, which approaches the common SNP-based heritability. Despite

relevant genes. In out-of-sample estimation and prediction, the 12,111 SNPs (or all SNPs in the HapMap 3 panel<sup>2</sup>) account for 40% (45%) of phenotypic variance in populations of European ancestry but only around 10–20% (14–24%) in populations of other ancestries. Effect sizes, associated regions and gene prioritization are similar

# A saturated map of common genetic variants associated with human height

**Figure 3. Prediction accuracy relative to European ancestry individuals across 17 quantitative traits and 5 continental populations in UKBB.**



# Multi-ancestry GWAS

# General analysis framework

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**Pooled  
analysis**

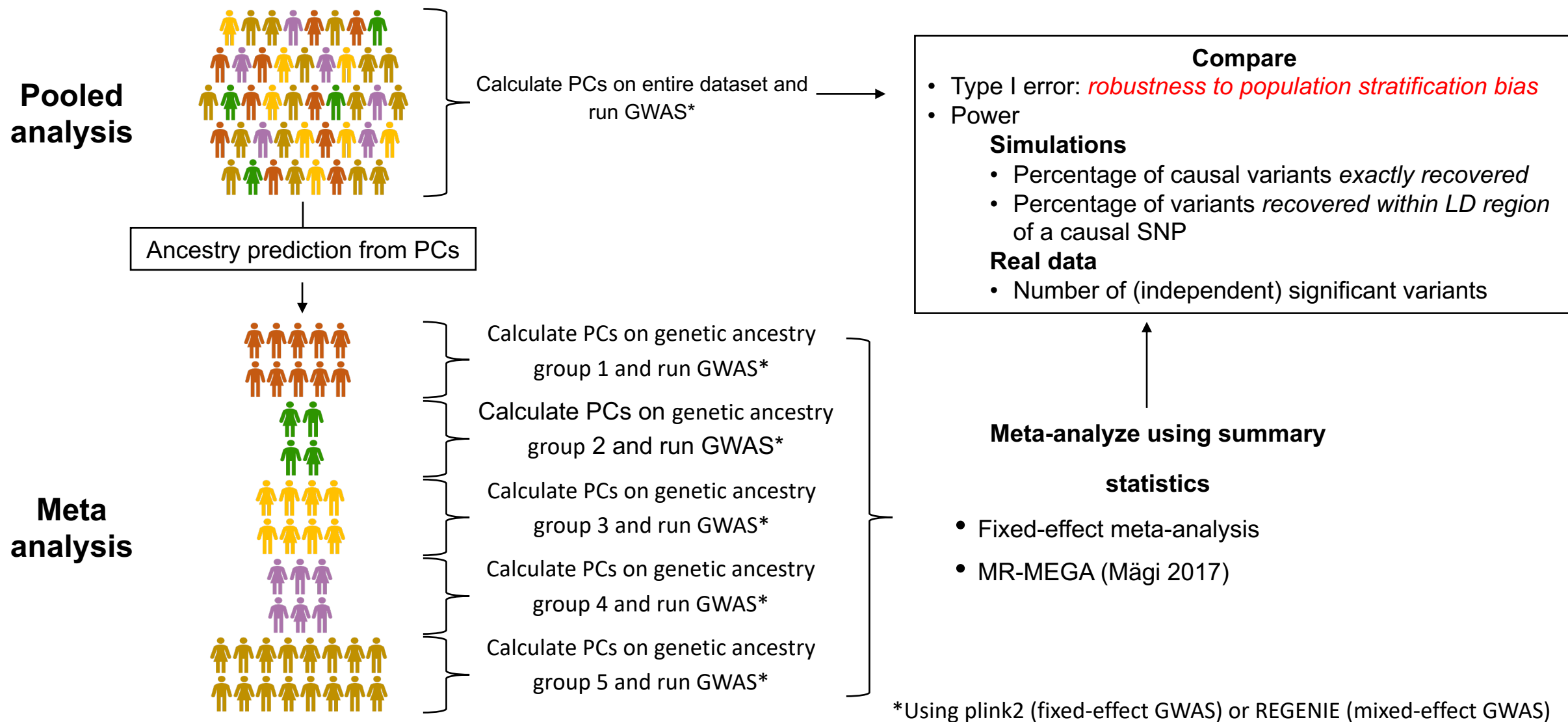


Construct genetic ancestry  
groups from genomic PCs

**Meta  
analysis**

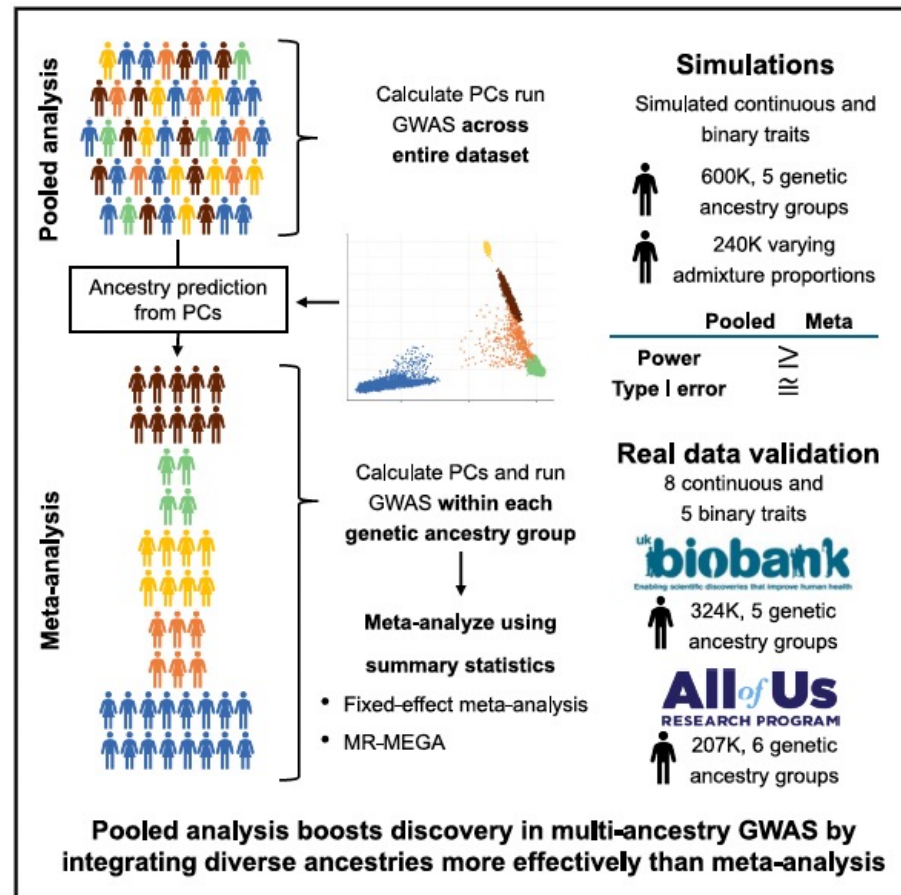


# General analysis framework



# Evaluating multi-ancestry genome-wide association methods: Statistical power, population structure, and practical implications

## Graphical abstract



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**Multi-ancestry GWASs enhance discovery in diverse populations, but optimal methods remain debated. Using theory, simulations, and analyses from the UK Biobank and All of Us, we show that pooled analysis outperforms meta-analysis in power while controlling for population structure, driven by allele-frequency differences across ancestries.**

# Annotating GWAS Results

# Annotating GWAS Results: Why It's Hard

## Going from association to mechanism is hard

- A GWAS hit flags a *region*, not which variant is causal or which gene it acts through
- The causal variant is typically buried among many others in LD (*fine-mapping/colocalization on session #4!*)
- Coding variants are relatively tractable: they change a protein directly, so the affected gene is usually clear

## Most (>90%) GWAS hits are regulatory, not coding

- Regulatory variants can act on genes far from the nearest gene, and their effects are often cell-type and context-specific

## But not impossible

- Several GWAS loci have already led to validated mechanisms and, in some cases, therapeutics (e.g., *IL23R/IL6R* pathway loci → biologics used in autoimmune disease)
- Annotation is how we start closing the gap. It doesn't prove causality, but it **generates and prioritizes hypotheses for functional follow-up**

# Annotation Strategies & Tools

**Positional / gene-based:** assign SNPs to nearby protein-coding genes

- ENSEMBL VEP, UCSC Genome Browser, ANNOVAR

**Regulatory:** check whether a variant sits in a known promoter/enhancer/accessible-chromatin region in a relevant cell type

- UCSC Genome Browser, HaploReg, RegulomeDB, FUMA

**Functional / molecular QTLs:** test association with gene expression (eQTL) or related molecular traits (sQTL, pQTL, mQTL) in relevant tissues

- GTEx Portal, FUMA

**LD / locus context:** see what else travels with your lead variant to understand the implicated region

- LDlink, LocusZoom

**Integrated platforms:** combine coding, regulatory, eQTL, conservation, and deleteriousness scores into one queryable resource

- FUMA, FAVOR, ENSEMBL, cS2G

# Basic (a) GWAS and (b) Meta-analysis Tutorial

*Open up HPC OnDemand!*

<https://hpc.nih.gov/ondemand/>